

Leonard Wong

AI Software Engineer | LLM Applications | Agentic Workflows

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SUMMARY

AI software engineer with 4+ years of public-sector software delivery experience, backed by earlier work in high-volume data platforms, APIs, and application security. Experienced in building LLM-powered applications and tool-using workflows with retrieval-augmented generation (RAG), function calling, and prompt engineering, with an emphasis on responsible AI, explainability, and secure delivery. Published researcher with experience in end-to-end delivery, UAT, incident management, and stakeholder enablement.

CORE SKILLS

AI & LLM: LLM application development, agentic workflows, RAG, tool use / function calling, prompt engineering, responsible AI, explainable AI

Engineering: Python, C#, JavaScript, SQL, ASP.NET, REST APIs, AWS DynamoDB, ETL, cloud-native design, scalable systems

Security & Delivery: OWASP ZAP, SAST/DAST, web application security, UAT, incident management, user training, product and platform strategy

Additional: HTML, CSS, PHP, Bootstrap, jQuery, JSON, data warehousing

PROFESSIONAL EXPERIENCE

NCS Pte. Ltd.

Jan 2022 - Present

AI Software Engineer

Apr 2026 - Present

- Develop LLM-powered applications and tool-using workflows incorporating RAG, function calling, and prompt engineering for secure, data-driven environments.
- Apply responsible and explainable AI practices alongside software security controls and operational support considerations for AI-enabled systems.
- Collaborate with stakeholders across requirements, validation, user enablement, and ongoing support to translate AI capabilities into practical software delivery.

Software Engineer

Jan 2022 - Mar 2026

- Owned end-to-end delivery of a confidential internal project, from implementation and deployment through post-release support.
- Planned and facilitated UAT by creating test scripts, coordinating stakeholder sessions, and driving issue resolution.
- Managed operational incidents and delivered user training and ongoing support to sustain adoption and service continuity.

Leadership: NCS Young Leaders; Brand Ambassador

Singapore Management University

Dec 2021

Research Assistant

- Evaluated 88 IS216 projects using OWASP ZAP and commercial DAST tools, complemented by static analysis across the same codebases.
- Contributed findings to peer-reviewed research on integrating application security into web programming education, published at ITiCSE 2022.

PROFESSIONAL EXPERIENCE (CONTINUED)

Alpha Ori Technologies

May - Jul 2020

Maritime Tech Solutions Intern

- Automated conversion of 150M+ rows containing 3B+ data points into AWS DynamoDB.
- Built API proof-of-concepts for three vendors to evaluate technical fit and adoption viability.
- Implemented data sorting and voyage analysis to identify operational trends.

OSM Thome (formerly Thome Ship Management)

May - Oct 2018

IT Intern

- Built an enterprise analytics portal presenting 15 key data points for decision-making.
- Delivered a ship-agency system supporting 160+ emails per hour.
- Integrated APIs to enrich vessel information across 230+ vessels.

ADDITIONAL EXPERIENCE

Rightspot | Intern - Resolved 20+ bugs within two weeks; built applicant-status and job-listing dashboards covering 250 applicants and 500+ companies.

Jul - Oct 2016

SamMobile | Contributing Writer (Freelance) - Published 1-2 articles per week for a 1.5M+ member audience, covering mobile software updates and product insights.

Jul - Dec 2013

EDUCATION

Singapore Management University - BSc Information Systems (Cyber Security)

2017 - 2022

Nanyang Polytechnic - Diploma in Business Informatics with Merit

2014 - 2017

Gachon University - Overseas Exchange

2015

PUBLICATION

XSS for the Masses: Integrating Security in a Web Programming Course using a Security Scanner - ITiCSE 2022, Apr 2022

Lwin Khin Shar, Kyong Jin Shim, Christopher M. Poskitt, and Li Ying Leonard Wong

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

AI Singapore - Certificate of Completion: LLM Application Developer Programme

Feb 2026

NUS-ISS - AI and Cybersecurity

Jun 2025

NUS-ISS - Explainable and Responsible Artificial Intelligence

May 2025

NUS-ISS - Graduate Certificate in Designing and Managing Products and Platforms

Mar 2025

NUS-ISS - Service Design

Jun 2024

NUS-ISS - Architecting Platforms as a Business

May 2024

NUS-ISS - Digital Product Strategy

May 2024

Singapore Computer Society - Certification in Managing Digital Products

Mar 2024

NUS-ISS - Managing Digital Products

Mar 2024

NUS-ISS - Graduate Certificate of Architecting Scalable Systems

Feb 2024

NUS-ISS - Platform Engineering

Dec 2022

NUS-ISS - Cloud Native Solution Design

Nov 2022

NUS-ISS - Architecting Software Solutions

Aug 2022

NUS-ISS - DevOps Engineering and Automation

Mar 2022

Scrum.org - Professional Scrum Master I (PSM I)

Feb 2022